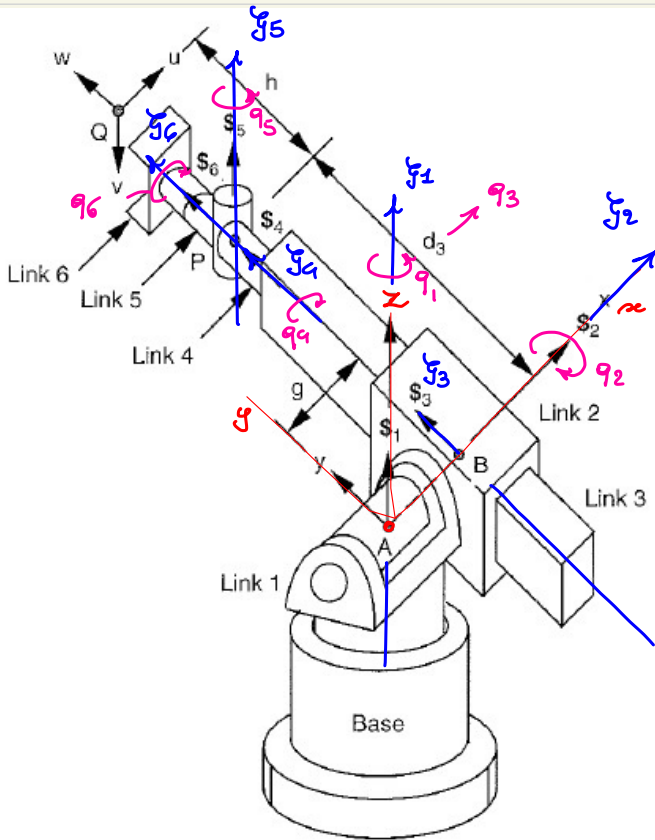


ESERCIZIO #1



Manipolatore di
Stanford

Sistema di rif. fisso $\{S\}$

$$\{S\} = \{A; x, y, z\}$$

Sistema di rif. end effector $\{B\}$

o) Parametrizzazione con
global POE,

Nella config. di rif

$$R_{Sb}(0) = \begin{Bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{Bmatrix}$$

$$P_{Sb}(0) = \{g \ h \ 0\}^T$$

Componenti dei twist
 ξ_i ($i=1, \dots, 6$) in forma
fissa: $\Upsilon_i = [\xi_i]^S$

$$\Upsilon_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix} \text{ (rotazionale); } \Upsilon_2 = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix}; \text{ (rotazionale)}$$

$$\Upsilon_3 = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}; \text{ (prismatica); } \Upsilon_4 = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix}; \text{ (rotazionale, polso)}$$

$$\Upsilon_5 = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix}; \text{ (rotazionale, polso)} \quad \Upsilon_6 = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix}; \text{ (rotazionale, polso)}$$

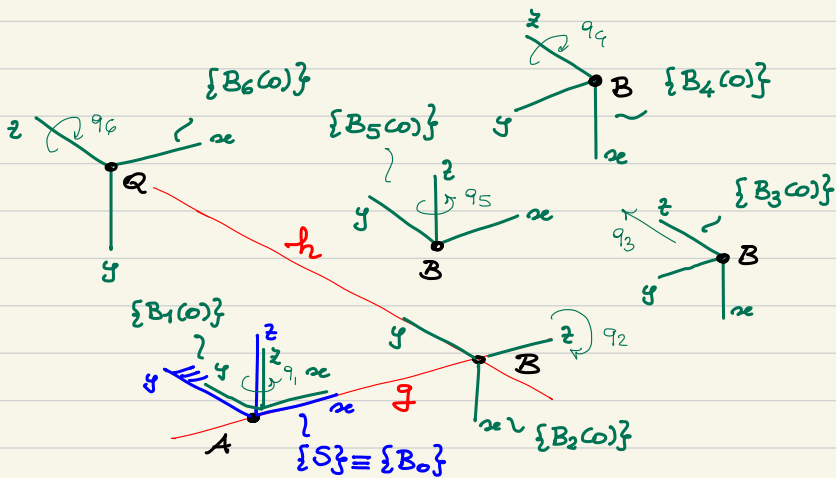
Ottenuti per ispezione data semplicità della config. di rif.

Dunque la struttura della parametrizzazione global DOE risulta:

$$g_{sb}(\underline{q}) = e^{\hat{Y}_1 q_1} e^{\hat{Y}_2 q_2} e^{\hat{Y}_3 q_3} \dots e^{\hat{Y}_6 q_6} g_{sb}(\omega),$$

$$\text{dove } \underline{q} = \{q_1 \ q_2 \ \dots \ q_6\}^T \text{ e } g_{sb}(\omega) = \begin{Bmatrix} R_{sb}(\omega) & p_{sb}(\omega) \\ 0^T & 1 \end{Bmatrix}$$

-) Occupiamoci adesso della parametrizzazione con local DOE:
Facciamo un semplice schema di riferimento:

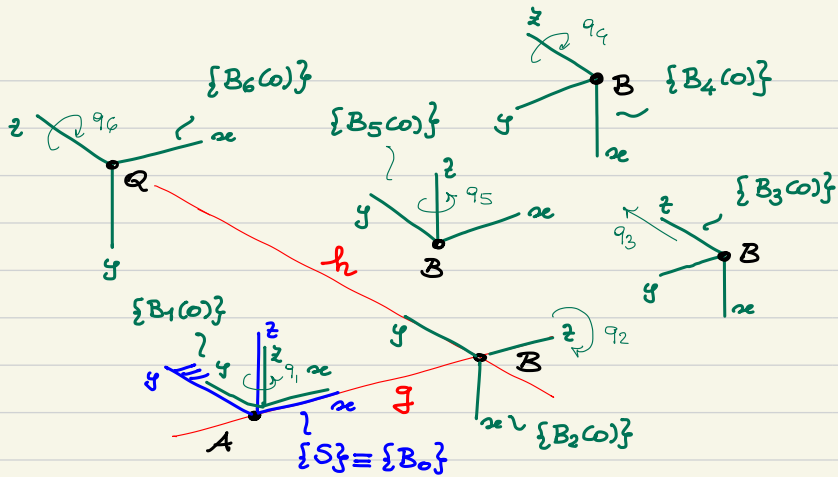


Avendo libertà di scelta i sistemi di rif. $\{B_1\}, \{B_2\}, \{B_3\}, \{B_4\}$ e $\{B_5\}$ vengono scelti con assi z allineati come gli assi dei giunti. Invece, $\{B_6\}$ è scelto coincidente alla terza end effector riportata nello schema originale.

Dunque, con la notazione $X_i = [f_i]^{b_i}$ e la struttura

$$g_{sb}(\underline{q}) = g_{0,1}(\omega) e^{\hat{X}_1 q_1} e^{\hat{X}_2 q_2} \dots e^{\hat{X}_6 q_6}$$

si ha:



$$\bullet) X_1 = \begin{Bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{Bmatrix}; \quad g_{0,1}(\omega) = I_{4 \times 4}$$

$$\bullet) X_2 = \begin{Bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{Bmatrix}; \quad g_{1,2}(\omega) = \begin{Bmatrix} R_y(\pi/2) \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix} \\ 0 \quad 0 \quad 0 \quad 1 \end{Bmatrix};$$

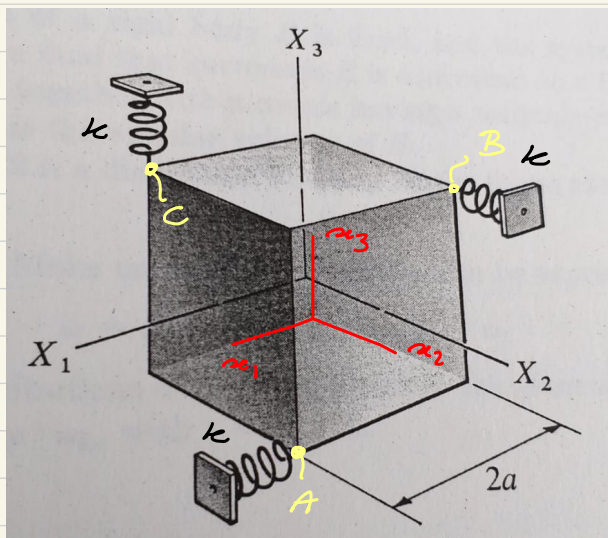
$$\bullet) X_3 = \begin{Bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \end{Bmatrix}; \quad g_{2,3}(\omega) = \begin{Bmatrix} R_x(-\pi/2) \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix} \\ 0 \quad 0 \quad 0 \quad 1 \end{Bmatrix};$$

$$\bullet) X_4 = \begin{Bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{Bmatrix}; \quad g_{3,4}(\omega) = I_{4 \times 4}$$

$$\bullet) X_5 = \begin{Bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{Bmatrix}; \quad g_{4,5}(\omega) = \begin{Bmatrix} R_x(\pi/2) R_y(-\pi/2) \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix} \\ 0 \quad 0 \quad 0 \quad 1 \end{Bmatrix};$$

$$\bullet) X_6 = \begin{Bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{Bmatrix}; \quad g_{5,6} = \begin{Bmatrix} R_x(-\pi/2) \begin{Bmatrix} 0 \\ \frac{1}{2} \\ 0 \\ 1 \end{Bmatrix} \\ 0 \quad 0 \quad 0 \quad 1 \end{Bmatrix}.$$

ESERCIZIO #2



Param. di $\{L\} = \{0; x_1, x_2, x_3\}$ rispetto a $\{S\} = \{0; X_1, X_2, X_3\}$ mediante $d_{00} = \{x \ y \ z\}^T$ e $\underline{q} = \{q_1 \ q_2 \ q_3\}^T$ della sequenza x_1, x_2, x_3 che corrisponde ad una 1-2-3, ossia $R_x(q_1) R_y(q_2) R_z(q_3)$.

IP. In config. rif. lunghezza a riposo delle molle è nulla e $0 \equiv 0$ e $q_1 = q_2 = q_3 = 0$.

Consideriamo che siano A, B e C in figura i vertici del cubo dove sono collegate i capi mobili delle molle.

$$\underline{s}_A = \underline{s}_0 + \underline{s}_\theta \times \underline{o}_A \quad \text{con } \underline{s}_\theta = \begin{Bmatrix} s_{q_1} \\ s_{q_2} \\ s_{q_3} \end{Bmatrix}; \quad \underline{o}_A \approx \begin{Bmatrix} 2a \\ 2a \\ 0 \end{Bmatrix}$$

$$\text{e } \underline{s}_0 = \begin{Bmatrix} s_u \\ s_v \\ s_z \end{Bmatrix}$$

Dunque esplicitamente:

$$\underline{\underline{S}}_A = \begin{Bmatrix} \delta x \\ \delta y \\ \delta z \end{Bmatrix} + \begin{Bmatrix} 0 & -\delta q_3 & \delta q_2 \\ \delta q_3 & 0 & -\delta q_1 \\ -\delta q_2 & \delta q_1 & 0 \end{Bmatrix} \begin{Bmatrix} 2a \\ 2a \\ 0 \end{Bmatrix} = \begin{Bmatrix} \delta x - 2a \delta q_3 \\ \delta y + 2a \delta q_3 \\ \delta z + 2a (\delta q_1 - \delta q_2) \end{Bmatrix}$$

$$\underline{\underline{S}}_B = \begin{Bmatrix} \delta x \\ \delta y \\ \delta z \end{Bmatrix} + \begin{Bmatrix} 0 & -\delta q_3 & \delta q_2 \\ \delta q_3 & 0 & -\delta q_1 \\ -\delta q_2 & \delta q_1 & 0 \end{Bmatrix} \begin{Bmatrix} 0 \\ 2a \\ 2a \end{Bmatrix} = \begin{Bmatrix} \delta x + 2a (\delta q_2 - \delta q_3) \\ \delta y - 2a \delta q_1 \\ \delta z + 2a \delta q_1 \end{Bmatrix}$$

$$\underline{\underline{S}}_C = \begin{Bmatrix} \delta x \\ \delta y \\ \delta z \end{Bmatrix} + \begin{Bmatrix} 0 & -\delta q_3 & \delta q_2 \\ \delta q_3 & 0 & -\delta q_1 \\ -\delta q_2 & \delta q_1 & 0 \end{Bmatrix} \begin{Bmatrix} 2a \\ 0 \\ 2a \end{Bmatrix} = \begin{Bmatrix} \delta x + 2a \delta q_2 \\ \delta y + 2a (\delta q_3 - \delta q_1) \\ \delta z - 2a \delta q_2 \end{Bmatrix}$$

L'energia potenziale elastica immagazzinata nelle molle ridotta:

$$U = \frac{1}{2} K [\underline{\underline{S}}_A^T \underline{\underline{S}}_A + \underline{\underline{S}}_B^T \underline{\underline{S}}_B + \underline{\underline{S}}_C^T \underline{\underline{S}}_C]$$

$$\underline{\underline{S}}_A = \underline{\underline{I}} \underline{\underline{S}}_0 + \begin{Bmatrix} 0 & 0 & -2a \\ 0 & 0 & 2a \\ 2a & -2a & 0 \end{Bmatrix} \underline{\underline{S}}_q = \underline{\underline{I}} \underline{\underline{S}}_0 + \underline{\underline{S}}_A \underline{\underline{S}}_q$$

$$\underline{\underline{S}}_B = \underline{\underline{I}} \underline{\underline{S}}_0 + \begin{Bmatrix} 0 & 2a & -2a \\ -2a & 0 & 0 \\ 2a & 0 & 0 \end{Bmatrix} \underline{\underline{S}}_q = \underline{\underline{I}} \underline{\underline{S}}_0 + \underline{\underline{S}}_B \underline{\underline{S}}_q$$

$$\underline{\underline{S}}_C = \underline{\underline{I}} \underline{\underline{S}}_0 + \begin{Bmatrix} 0 & 2a & 0 \\ -2a & 0 & 2a \\ 0 & -2a & 0 \end{Bmatrix} \underline{\underline{S}}_q = \underline{\underline{I}} \underline{\underline{S}}_0 + \underline{\underline{S}}_C \underline{\underline{S}}_q$$

Dunque

$$\underline{\underline{S}}_A = [\underline{\underline{I}} \quad \underline{\underline{S}}_A] \begin{Bmatrix} \underline{\underline{S}}_0 \\ \underline{\underline{S}}_q \end{Bmatrix}; \quad \underline{\underline{S}}_A^T = \begin{Bmatrix} \underline{\underline{S}}_0^T & \underline{\underline{S}}_q^T \end{Bmatrix} \begin{bmatrix} \underline{\underline{I}} \\ \underline{\underline{S}}_A^T \end{bmatrix}$$

$$\begin{aligned} \text{Quindi } \underline{\underline{S}}_A^T \underline{\underline{S}}_A &= \begin{Bmatrix} \underline{\underline{S}}_0^T & \underline{\underline{S}}_q^T \end{Bmatrix} \begin{bmatrix} \underline{\underline{I}} \\ \underline{\underline{S}}_A^T \end{bmatrix} [\underline{\underline{I}} \quad \underline{\underline{S}}_A] \begin{Bmatrix} \underline{\underline{S}}_0 \\ \underline{\underline{S}}_q \end{Bmatrix} \\ &= \begin{Bmatrix} \underline{\underline{S}}_0^T & \underline{\underline{S}}_q^T \end{Bmatrix} \begin{Bmatrix} \underline{\underline{I}} & \underline{\underline{S}}_A \\ \underline{\underline{S}}_A^T & \underline{\underline{S}}_A^T \underline{\underline{S}}_A \end{Bmatrix} \begin{Bmatrix} \underline{\underline{S}}_0 \\ \underline{\underline{S}}_q \end{Bmatrix} \end{aligned}$$

In tal modo si ottiene:

$$U = \frac{1}{2} k \{ \underline{\delta o}^T \quad \underline{\delta q}^T \} \begin{Bmatrix} 3I & S_A + S_B + S_C \\ S_A^T + S_B^T + S_C^T & S_A^T S_A + S_B^T S_B + S_C^T S_C \end{Bmatrix} \begin{Bmatrix} \underline{\delta o} \\ \underline{\delta q} \end{Bmatrix}$$

↙ esplicitam.

$$\begin{pmatrix} 3 & 0 & 0 & 0 & 4a & -4a \\ 0 & 3 & 0 & -4a & 0 & 4a \\ 0 & 0 & 3 & 4a & -4a & 0 \\ 0 & -4a & 4a & 16a^2 & -4a^2 & -4a^2 \\ 4a & 0 & -4a & -4a^2 & 16a^2 & -4a^2 \\ -4a & 4a & 0 & -4a^2 & -4a^2 & 16a^2 \end{pmatrix}$$

La linearizzazione effettuata sugli spostamenti causati dalle rotazioni ha consentito di tenere solo termini lineari che hanno dato luogo, nella espressione della U , solo a termini al più quadratici.

Questo consente di scrivere la U come una forma quadratica la cui espressione compatta è stata sistematicamente ricavata sopra.